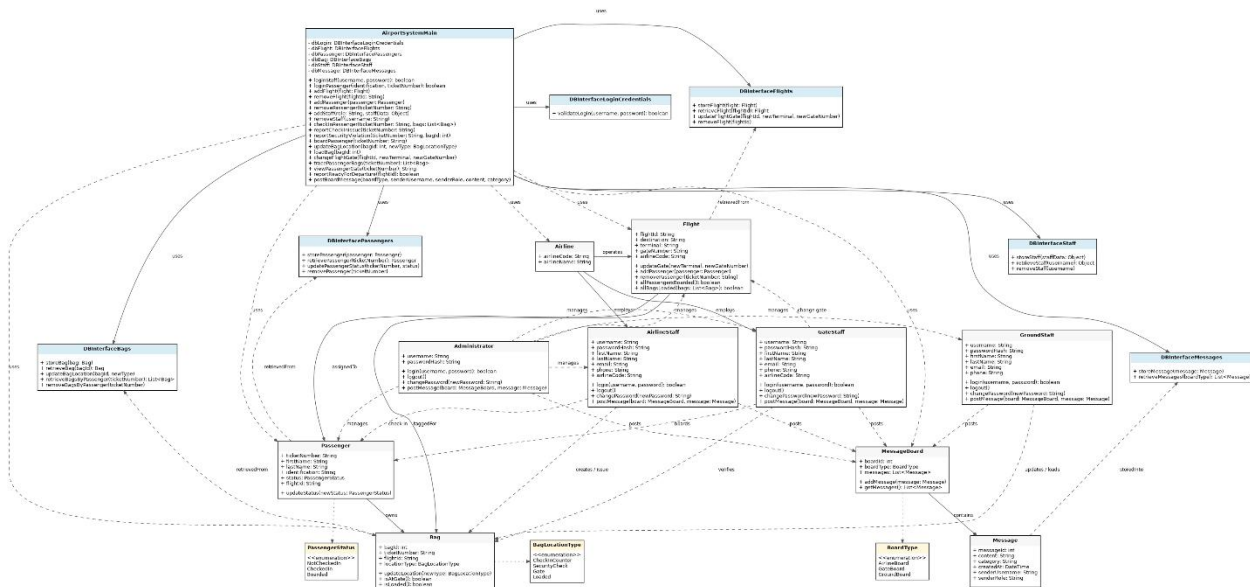


CS 5336 - Web Application Development

Project Title: Luggage Handling App

Application Design Document

Class Diagram



Detailed Description

Common to all classes

All classes have a constructor. All classes will also have getter methods for their attributes. Because all classes contain these functions, they are not included in the method descriptions below.

Classes

Classname: AirportSystemMain

Attributes:

- dbLogin: DBInterfaceLoginCredentials
- dbFlight: DBInterfaceFlights
- dbPassenger: DBInterfacePassengers
- dbBag: DBInterfaceBags

- dbStaff: DBInterfaceStaff
- dbMessage: DBInterfaceMessages

Methods:

- loginStaff(username: String, password: String): boolean
- loginPassenger(identification: String, ticketNumber: String): boolean
- logout(userId: String)
- addFlight(flight: Flight)
- removeFlight(flightId: String)
- addPassenger(passenger: Passenger)
- removePassenger(ticketNumber: String)
- addStaff(role: String, staffData: Object)
- removeStaff(username: String)
- checkInPassenger(ticketNumber: String, bags: List<Bag>)
- reportCheckInIssue(ticketNumber: String)
- reportSecurityViolation(ticketNumber: String, bagId: int)
- boardPassenger(ticketNumber: String)
- updateBagLocation(bagId: int, newType: BagLocationType)
- loadBag(bagId: int)
- changeFlightGate(flightId: String, newTerminal: String, newGateNumber: String)
- tracePassengerBags(ticketNumber: String): List<Bag>
- viewPassengerGate(ticketNumber: String): String
- reportReadyForDeparture(flightId: String): boolean
- postBoardMessage(boardType: BoardType, senderUsername: String, senderRole: String, content: String, category: String)

Classname: Airline

Attributes:

- airlineCode: String
- airlineName: String

Methods:

- None from the design view.

Classname: Administrator

Attributes:

- username: String
- passwordHash: String

Methods:

- login(username: String, password: String): boolean

- logout()
- changePassword(newPassword: String)
- postMessage(board: MessageBoard, message: Message)

Classname: GroundStaff

Attributes:

- username: String
- passwordHash: String
- firstName: String
- lastName: String
- email: String
- phone: String

Methods:

- login(username: String, password: String): boolean
- logout()
- changePassword(newPassword: String)
- postMessage(board: MessageBoard, message: Message)

Classname: GateStaff

Attributes:

- username: String
- passwordHash: String
- firstName: String
- lastName: String
- email: String
- phone: String
- airlineCode: String

Methods:

- login(username: String, password: String): boolean
- logout()
- changePassword(newPassword: String)
- postMessage(board: MessageBoard, message: Message)

Classname: AirlineStaff

Attributes:

- username: String
- passwordHash: String
- firstName: String
- lastName: String
- email: String
- phone: String
- airlineCode: String

Methods:

- login(username: String, password: String): boolean
- logout()
- changePassword(newPassword: String)
- postMessage(board: MessageBoard, message: Message)

Classname: Flight

Attributes:

- flightId: String
- destination: String
- terminal: String
- gateNumber: String
- airlineCode: String

Methods:

- updateGate(newTerminal: String, newGateNumber: String)
- addPassenger(passenger: Passenger)
- removePassenger(ticketNumber: String)
- allPassengersBoarded(): boolean
- allBagsLoaded(bags: List<Bag>): boolean

Classname: Passenger

Attributes:

- ticketNumber: String
- firstName: String
- lastName: String
- identification: String
- status: PassengerStatus
- flightId: String

Methods:

- updateStatus(newStatus: PassengerStatus)

Classname: Bag

Attributes:

- bagId: int
- ticketNumber: String
- flightId: String
- locationType: BagLocationType

Methods:

- updateLocation(newType: BagLocationType)
- isAtGate(): boolean
- isLoaded(): boolean

Classname: MessageBoard

Attributes:

- boardId: int
- boardType: BoardType
- messages: List<Message>

Methods:

- addMessage(message: Message)
- getMessages(): List<Message>

Classname: Message

Attributes:

- messageId: int
- content: String
- category: String
- createdAt: DateTime
- senderUsername: String
- senderRole: String

Methods:

- None from the design view.

Supporting Database Interface Classes

Classname: DBInterfaceLoginCredentials

Attributes:

- None

Methods:

- validateLogin(username: String, password: String): boolean

Classname: DBInterfaceFlights

Attributes:

- None

Methods:

- storeFlight(flight: Flight)

- retrieveFlight(flightId: String): Flight

- updateFlightGate(flightId: String, newTerminal: String, newGateNumber: String)

- removeFlight(flightId: String)

Classname: DBInterfacePassengers

Attributes:

- None

Methods:

- storePassenger(passenger: Passenger)

- retrievePassenger(ticketNumber: String): Passenger

- updatePassengerStatus(ticketNumber: String, status: PassengerStatus)

- removePassenger(ticketNumber: String)

Classname: DBInterfaceBags

Attributes:

- None

Methods:

- storeBag(bag: Bag)

- retrieveBag(bagId: int): Bag

- updateBagLocation(bagId: int, newType: BagLocationType)

- retrieveBagsByPassenger(ticketNumber: String): List<Bag>

- removeBagsByPassenger(ticketNumber: String)

Classname: DBInterfaceStaff

Attributes:

- None

Methods:

- storeStaff(staffData: Object)

- retrieveStaff(username: String): Object

- removeStaff(username: String)

Classname: DBInterfaceMessages

Attributes:

- None

Methods:

- storeMessage(message: Message)
- retrieveMessages(boardType: BoardType): List<Message>

Enumeration Types

Type: PassengerStatus

- NotCheckedIn
- CheckedIn
- Boarded

Type: BagLocationType

- CheckInCounter
- SecurityCheck
- Gate
- Loaded

Type: BoardType

- AirlineBoard
- GateBoard
- GroundBoard